

Prescribing the ice–regolith contact angle

Wall free energy, its weak form, and why no surface energy survives into the residual coefficient

lunar_regolith_DSM

2026-09-16

Summary. The substrate in this two-phase model is the domain wall, not a phase field, so it must act through a boundary condition. Adding Cahn’s wall free energy $f_w(\phi) = \gamma_{as} + (\gamma_{is} - \gamma_{as})h(\phi)$ to the functional supplies a natural condition on $\partial_n \phi$ that fixes the contact angle. Carrying it through the weak form gives a residual term $-3M \cos \theta h'(\phi)/6$, with no ϵ and no surface energy in the coefficient. Both cancellations are derived below. The γ_{ia} that appears in $\cos \theta$ is *not* the one that cancels.

1. What the wall energy is

Let ϕ be the ice fraction, $\phi = 1$ ice and $\phi = 0$ vapour, and let Γ be the part of $\partial\Omega$ that represents regolith. The wall carries an energy per unit area that interpolates between its two dry states,

$$f_w(\phi) = \gamma_{as}(1 - h(\phi)) + \gamma_{is}h(\phi) = \gamma_{as} + (\gamma_{is} - \gamma_{as})h(\phi), \quad h(\phi) = \phi^2(3 - 2\phi), \quad (1)$$

so $f_w(0) = \gamma_{as}$ (wall against vapour) and $f_w(1) = \gamma_{is}$ (wall against ice).

Two properties of h matter. $h(0) = 0$ and $h(1) = 1$ give the correct end states. $h'(0) = h'(1) = 0$ makes the term inert in both bulk phases: it can only act where the ice–vapour interface actually meets the wall, and cannot shift the bulk equilibria of the double well. A linear interpolant would violate the second property and bias ϕ everywhere along Γ .

Note that (1) contains no γ_{ia} . Only the two substrate energies appear. This is worth holding onto; γ_{ia} enters later, from a different direction, and that is the whole reason $\cos \theta$ emerges.

2. The functional the solver actually evolves

The ice residual implemented in `assembly.c` is

$$R = N \phi_t + 3M\epsilon \nabla N \cdot \nabla \phi + \frac{3M}{\epsilon} f_1(\phi) N - (\text{source}), \quad f_1 = W'(\phi), \quad (2)$$

with $M = \text{mob_sub}$ and the deliberately half-normalised double well

$$W(\phi) = \frac{1}{2} \phi^2(1 - \phi)^2. \quad (3)$$

Equation (2) is the weak form of the gradient flow

$$\frac{\partial \phi}{\partial t} = -\frac{3M}{\epsilon} \frac{\delta \tilde{\mathcal{F}}}{\delta \phi}, \quad \tilde{\mathcal{F}}[\phi] = \int_{\Omega} \left[\frac{\epsilon^2}{2} |\nabla \phi|^2 + W(\phi) \right] d\Omega. \quad (4)$$

$\tilde{\mathcal{F}}$ is a *shape* functional. It carries no surface energy: there is no γ anywhere in (4), and none in (2). In this solver γ_{ia} reaches the dynamics only through the kinetic calibration $d_{0,\text{sub}0} \rightarrow \lambda_{\text{sub}} \rightarrow \tau_{\text{sub}} \rightarrow M, \alpha_{\text{sub}}$, never as an energy scale multiplying the PDE. A physical energy therefore cannot be added to (4) directly. It has to be converted first, and the conversion factor is where the interesting algebra lives.

2.1 The conversion factor, and where 1/6 comes from

For the equilibrium profile of (3),

$$\phi(x) = \frac{1}{2} \left(1 + \tanh \frac{x}{2\epsilon} \right), \quad \text{so} \quad \phi' = \frac{\phi(1-\phi)}{\epsilon}, \quad (5)$$

the first integral of $\epsilon^2 \phi'' = W'(\phi)$ is $\frac{\epsilon^2}{2} \phi'^2 = W$, i.e. the two terms of the integrand are equal (equipartition). Hence the interfacial excess of $\tilde{\mathcal{F}}$ per unit area is

$$\int_{-\infty}^{\infty} \left[\frac{\epsilon^2}{2} \phi'^2 + W \right] dx = \epsilon^2 \int \phi'^2 dx = \frac{\epsilon^2}{16\epsilon^2} \int \text{sech}^4 \frac{x}{2\epsilon} dx = \frac{1}{16} \cdot 2\epsilon \cdot \frac{4}{3} = \boxed{\frac{\epsilon}{6}}. \quad (6)$$

(Verified numerically to eight digits.) The same interface carries physical energy γ_{ia} per unit area. The two functionals are therefore related by a pure number times γ_{ia}/ϵ ,

$$\mathcal{F}_{\text{phys}} = \frac{6\gamma_{ia}}{\epsilon} \tilde{\mathcal{F}} \quad \Longleftrightarrow \quad \text{a physical energy } E \text{ enters } \tilde{\mathcal{F}} \text{ as } \tilde{E} = \frac{\epsilon}{6\gamma_{ia}} E. \quad (7)$$

3. The weak form, in full

Write the total functional including the wall term,

$$\tilde{\mathcal{F}}_{\text{tot}}[\phi] = \int_{\Omega} \left[\frac{\epsilon^2}{2} |\nabla \phi|^2 + W(\phi) \right] d\Omega + \oint_{\Gamma} \tilde{f}_w(\phi) d\Gamma, \quad \tilde{f}_w = \frac{\epsilon}{6\gamma_{ia}} f_w. \quad (8)$$

Take the first variation with an arbitrary test function η :

$$\delta \tilde{\mathcal{F}}_{\text{tot}}[\eta] = \int_{\Omega} \left[\epsilon^2 \nabla \phi \cdot \nabla \eta + W'(\phi) \eta \right] d\Omega + \oint_{\Gamma} \tilde{f}_w'(\phi) \eta d\Gamma. \quad (9)$$

3.1 Strong form and the natural boundary condition

To read off the boundary condition, integrate the gradient term by parts:

$$\int_{\Omega} \epsilon^2 \nabla \phi \cdot \nabla \eta d\Omega = - \int_{\Omega} \epsilon^2 \nabla^2 \phi \eta d\Omega + \oint_{\partial\Omega} \epsilon^2 (\nabla \phi \cdot \mathbf{n}) \eta dS. \quad (10)$$

Substituting into (9) and collecting,

$$\delta \tilde{\mathcal{F}}_{\text{tot}}[\eta] = \int_{\Omega} \left[-\epsilon^2 \nabla^2 \phi + W'(\phi) \right] \eta d\Omega + \oint_{\partial\Omega} \left[\epsilon^2 \partial_n \phi + \tilde{f}_w'(\phi) \right] \eta dS. \quad (11)$$

Because η is arbitrary in Ω and independently arbitrary on $\partial\Omega$, stationarity requires both brackets to vanish. The second is the natural boundary condition,

$$\epsilon^2 \partial_n \phi + \tilde{f}_w'(\phi) = 0. \quad (12)$$

On any face where no wall term is declared, $\tilde{f}_w' \equiv 0$ and (12) reduces to $\partial_n \phi = 0$: the zero-flux condition the solver had everywhere before this feature, which is a 90° contact angle imposed by omission rather than by choice.

3.2 Evaluating $\tilde{f}_w'$, and the appearance of $\cos \theta$

From (1) and (7),

$$\tilde{f}_w'(\phi) = \frac{\epsilon}{6\gamma_{ia}} (\gamma_{is} - \gamma_{as}) h'(\phi) = -\frac{\epsilon}{6} \underbrace{\frac{\gamma_{as} - \gamma_{is}}{\gamma_{ia}}}_{\cos \theta} h'(\phi) = -\frac{\epsilon}{6} \cos \theta h'(\phi). \quad (13)$$

This is the step worth pausing on. **Young's equation was not assumed.** The wall energy (1) contains only γ_{as} and γ_{is} ; γ_{ia} entered solely through the normalisation (7) of the *bulk* functional. Their ratio is $(\gamma_{as} - \gamma_{is})/\gamma_{ia}$, which *is* Young's $\cos \theta$. The non-dimensionalisation produces it.

3.3 The contact angle this enforces

Insert (13) into (12):

$$\epsilon^2 \partial_n \phi = \frac{\epsilon}{6} \cos \theta h'(\phi) = \epsilon \cos \theta \phi(1 - \phi) \quad \implies \quad \partial_n \phi = \cos \theta \frac{\phi(1 - \phi)}{\epsilon}, \quad (14)$$

using $h'(\phi)/6 = \phi(1 - \phi)$. By (5) the right-hand side is $\cos \theta |\nabla \phi|$, and writing $\nabla \phi = |\nabla \phi| \mathbf{m}$ with $\mathbf{m}$ the interface normal pointing into the ice,

$$\boxed{\mathbf{m} \cdot \mathbf{n} = \cos \theta} \quad (15)$$

with $\mathbf{n}$ the outward wall normal: the contact angle measured through the ice. Limits check: $\cos \theta = +1$ puts $\mathbf{m} = \mathbf{n}$, the interface normal pointing straight out of the wall, i.e. ice sheeted flat onto it, $\theta = 0$; $\cos \theta = -1$ gives $\theta = 180^\circ$, a point contact.

4. The residual and Jacobian terms

The Galerkin form does *not* integrate the gradient term by parts — the integration by parts above was only to expose the boundary condition. The weak residual is (9) with $\eta = N_a$, multiplied by the prefactor $3M/\epsilon$ of the gradient flow (4):

$$R_a = \int_{\Omega} N_a \phi_t d\Omega + \frac{3M}{\epsilon} \left\{ \int_{\Omega} [\epsilon^2 \nabla N_a \cdot \nabla \phi + W' N_a] d\Omega + \oint_{\Gamma} \tilde{f}_w'(\phi) N_a d\Gamma \right\} - (\text{source}). \quad (16)$$

The two volume terms reproduce $3M\epsilon$ and $3M/\epsilon$ in (2) exactly. The boundary term gives, using (13),

$$R_a^{\Gamma} = \frac{3M}{\epsilon} \left(-\frac{\epsilon}{6} \cos \theta h'(\phi) \right) N_a = \boxed{-3M \cos \theta \frac{h'(\phi)}{6} N_a} \quad (17)$$

and, since the wall energy has no ϕ_t dependence, a Jacobian with no shift term,

$$J_{ab}^{\Gamma} = \frac{\partial R_a^{\Gamma}}{\partial \phi_b} = -3M \cos \theta \frac{h''(\phi)}{6} N_a N_b, \quad h''(\phi) = 6(1 - 2\phi). \quad (18)$$

Equation (17) is `assembly.c` verbatim: `Rb[a][0] = -rwb * 3.0 * mob_b * costhet * (dh/6.0) * N0b[a].`

4.1 Where each factor came from

factor	origin	equation
$3M$	the gradient-flow prefactor $3M/\epsilon$, as in the interior form	(4)
$1/6$	the interfacial excess $\epsilon/6$ of the dimensionless $\tilde{\mathcal{F}}$	(6)
$\cos \theta$	$(\gamma_{as} - \gamma_{is})/\gamma_{ia}$, produced by the normalisation	(13)
no ϵ	cancels between $\epsilon/6$ and $3M/\epsilon$	(17)
no γ prefactor	γ_{ia} survives only inside $\cos \theta$	(13)

4.2 Why one cannot simply “keep the γ_{ia} ”

A natural objection: the hand derivation gives $\delta f_w / \delta \phi = -\gamma_{ia} \cos \theta h'$, so why does γ_{ia} not multiply the residual?

Because that expression already contains γ_{ia} twice in disguise. Writing $f_w = \gamma_{as} - \gamma_{ia} \cos \theta h$ substitutes Young’s equation into (1); the γ_{ia} introduced there is cancelled by the γ_{ia} in $\cos \theta$, leaving $(\gamma_{is} - \gamma_{as})h$, which is (1) again. The *only* γ_{ia} with independent standing is the one in the normalisation (7), and it is that one which converts $(\gamma_{as} - \gamma_{is})$ into $\cos \theta$.

There is a real physical assumption buried here, and it is worth naming: the γ_{ia} in (7) is the ice–vapour energy the *bulk* model realises, while the γ_{ia} one might write in f_w is the ice–vapour energy at the *contact line*. They are the same interface, so they are the same number, and the ratio is one. If they were allowed to differ — as a sensitivity test, say — the ratio would survive and γ_{ia} would reappear explicitly.

4.3 Independent check

demo/Metamorph.c in the same PetIGA tree carries a contact-angle term derived by a different route, `R[a][0] = -N0[a]*3.0*eps*mob*|grad phi|*costhet`. Since $\epsilon|\nabla\phi| = \phi(1 - \phi) = h'(\phi)/6$ on the equilibrium profile (5), that is (17) exactly. Two derivations, one coefficient.

The $h(\phi)$ form is nonetheless preferable: it needs no $1/|\nabla\phi|$ regularisation (Metamorph clamps $|\nabla\phi|$ at 10^{-5} to avoid dividing by zero in its Jacobian), its Jacobian is an exact mass-matrix block, and it remains correct away from the equilibrium profile, where $\epsilon|\nabla\phi| \neq \phi(1 - \phi)$.

5. Clamping, and why it is required

$h'(\phi) = 6\phi(1 - \phi)$ changes sign for $\phi < 0$ and $\phi > 1$. So does (17), and since $R = N\phi_t + \dots = 0$, a sign-flipped wall term drives ϕ *further* out of range: a runaway.

Nothing in the bulk opposes it. The interior form evaluates f_1 and the localiser at the clamped $\phi_c \in [0, 1]$, so for $\phi < 0$ both are identically zero — there is no restoring force outside the range by construction. An unclamped wall term is then the only term acting there, and it acts outward.

This is not hypothetical. At $\epsilon/R = 1/100$ a -2×10^{-6} excursion at the wall grew to -0.05 within a few hundred steps, crossed `-phase_lo`, and the residual’s domain guard then returned a zero residual, which SNES read as $\|F\| = 0$ and reported as converged. The solution froze for the remaining 23,370 steps while the clock ran on to t_{final} and the run reported success. The implementation therefore clamps ϕ to $[0, 1]$ before evaluating h' , with the Jacobian clamping identically and carrying the chain-rule factor $d\phi_c/d\phi$, so that it remains an exact Jacobian of the clamped residual.

6. Using it

<code>-wall_faces</code>	<code>"y0,y1"</code>	which domain faces are regolith
<code>-gamma_is</code>		ice-regolith surface energy [J/m ²]
<code>-gamma_as</code>		air-regolith surface energy [J/m ²]
<code>-gamma_ia</code>		ice-air; default $(\Sigma_i + \Sigma_a)/2$

The solver forms $\cos \theta = (\gamma_{as} - \gamma_{is})/\gamma_{ia}$, refuses the run if $|\gamma_{as} - \gamma_{is}| > \gamma_{ia}$ (no angle exists; one phase wets completely), and prints both the energies and the derived θ .

On the default. γ_{ia} defaults to $(\Sigma_i + \Sigma_a)/2$, not to Σ_i . In the convention these projects inherit, $\Sigma_i = \gamma_{ia} + \gamma_{is} - \gamma_{as}$ and $\Sigma_a = \gamma_{ia} + \gamma_{as} - \gamma_{is}$, so $\Sigma_i + \Sigma_a = 2\gamma_{ia}$ and the ice-air energy is their mean — which is also what the solver’s own double-well coefficient $C = (\Sigma_i + \Sigma_a)/2 + \Lambda$ says. Σ_i alone equals γ_{ia} only when $\Sigma_a = \Sigma_i$, which `enceladus_DSM` enforces and `lunar_regolith_DSM` does not; with the lunar defaults (0.109, 0.132) taking Σ_i understates γ_{ia} by 10.6%, and that propagates directly into θ . Pass `-gamma_ia` explicitly whenever the substrate matters.

The only admissibility constraint is $|\gamma_{as} - \gamma_{is}| \leq \gamma_{ia}$. The three-phase triple-well constraint $\gamma_{is} > \frac{\gamma_{ia}}{2}(1 - \cos \theta)$ does *not* apply: it came from requiring a positive Σ_s , and there is no sediment phase and no triple well here.

7. Validation

Five prescribed angles from 30° to 150° in a flat channel recover Young’s prediction with a maximum error of 0.127° and an RMS of 0.062°, against a 2° acceptance criterion, with the error falling as ϵ/R is refined. A droplet on a single wall, which has no confinement, agrees with the channel, so the angle is set by the boundary condition rather than by the geometry. Details in `studies/contact.angle/`.